Answer to evaluate 1:

Calculate $[H^+]$ from a known pH. Find $[H^+]$ if pH = 8.5

$[H^+] = 10^{-pH}$

$[H^+] = 10^{-8.5}$

$[H^+] = 3.2 \times 10^{-9}$ M

Answer to Evaluate 2:

Calculate the pH of a 2 L solution containing 10 mL of 5 M acetic acid and 10 mL of 1 M sodium acetate. The pK_a for acetic acid is 4.76.

Answer:

First, calculate the concentrations of the acid and conjugate base, expressing all concentrations in units of moles per liter (Molar Concentration).

Acetic acid: (0.01 L)(5 M)/(2 L) = 0.025 M

Sodium acetate: (0.01 L) (1M)/ (2 L)= 0.005 M

Substitute the concentrations of the acid and conjugate base into the Henderson–Hasselbalch equation.

pH = pK_a + log ([acetate] / [acetic acid])

pH = 4.76 + log (0.005 / 0.025)

pH = 4 .76 - 0.70

pH = 4 .06

Answer to Evaluate 3:

The pK_a of the phosphate buffer system ($[H_2PO_4^-]/[HPO_4^{2-}]$) is 6.8. What are the relative concentrations of $[H_2PO_4^-]$ and $[HPO_4^{2-}]$ in a urine sample that has a pH of 4.8.

$$pH = pK_a + \log \frac{H_2PO_4^-}{[HPO_4^{2-}]} \qquad 1.13$$

Substitute the pH and pK_a values into the Henderson–Hasselbalch equation.

$$4.8 = 6.8 + \log \frac{H_2PO_4^-}{[HPO_4^{2-}]} \qquad 1.14$$

Rearrange the equation 1.14

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